JOSEFFUS SANTOS

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EDUCATION

University of Manitoba — Winnipeg, MB Canada
Bachelor of Science Honours Co-op Computer Science DGPA 3.9 / 4.0 — Expected Aug 2024
Bachelor of Science Major Biochemistry — Graduated

RELEVANT TECHNICAL SKILLS

- JavaScript
- Node.js
- Mocha / Jest
- React
- Bootstrap
- Ubuntu
- TypeScript
- Python
- PyTest
- OpenCV
- GeoPandas
- Numpy
- Scikit
- Tensorflow 2
- Java
- JUnit / Espresso
- Mockito
- Android Java
- C / C++
- Catch2
- C#
- SQL
- Blender
- KiCAD
- AWS

RELEVANT EXPERIENCE

Software Developer, Co-op (Full Stack) — Jan 2023 – Apr 2023
AGCO Corporation

- Implemented new functional React components to display remote sensor data from LoRaWAN endpoints on a dashboard. Converted vanilla JavaScript class based React components to TypeScript functional React components.
- Contributed new backend features in Python to support new API functionality using Chalice/Flask for serverless AWS applications running in Docker containers. Implemented Python classes to support changes in LoRaWAN endpoint firmware.
- Managed local dev environments in Docker containers to run CI/CD pipelines locally.
- Created PyTest and Jest, unit and integration tests to validate code functionality and verify implementation.
- Participated in Scrum activities: code reviews, standups, retrospectives, sprint planning.

Research Technician — Feb 2011 - Present
Canadian Grain Commission, Government of Canada

- Created an ETL pipeline in Python to scrape weather data and collect the information in a PostGres DB. Created dockerfiles to build custom development environment based on Nvidia docker images. Used scikit-learn and Tensorflow to explore ML/AI models for the prediction of crop quality from weather data.
- Developed programs in C# to extract data and add custom functions to instrument control software to speed up data analysis.
- Developed VBA scripts used across Canada to automate data handling in Excel for the evaluation of research data.
- AWS cloud administration. Designed and maintained a network of Windows and Linux EC2 instances to trial data analysis on the cloud with deployment scripts for users to automate ssh tunneling, instance start/stop, RDP connection, and deployment of Jupyter notebook servers.
- Designed and fabricated an Atmel SAMD based robotic liquid sampling and dispensing system with firmware written in C++
- Gathered software requirements from stakeholders internal and external to the organization. Implemented new features based on user feedback.
- Contributed to data collection, analysis, and authoring research papers. Conducted journal searches and wrote review summaries of the current state of research on topics of interest.
- Planned research projects. Occasionally supervised and allocated tasks for teams of 4-15 students and technicians.

PROJECTS

Personal cloud / Home Lab | ZFS, Linux, GCP, AWS, Virtualbox, Docker, HTML, CSS, JavaScript — 2015 - Present

- Network of multiple LAN hosted, cloud hosted, and hypervisor hosted web services
- Mixture of LAN Linux and Windows hosts; Linux hosts on AWS that run applications in docker to provide VPN, Reverse Proxy, SOCKS5, SSH, remote desktop, content delivery, server monitoring and other services.

Forest Fire Prediction Model | Python, Docker, PostGres, QGIS, GeoPandas, sklearn, Tensorflow — 2022

- Used Python to scrape data from weather stations nearest forest fires reported in Saskatchewan in the past 20 years
- Extended PostGres and Nvidia Tensorflow containers to build dev environments for machine learning
- Trained and evaluated SVM and Random Forest models for forest fire severity prediction using weather data

Meal Planner Calorie Tracking App | Android, Java, JUnit, Espresso, HSQLDB — 2022

- With a team of 5, designed and built a Java Android app following Agile development practices
- Applied software design patterns, implemented object oriented class design, and thorough unit testing
- Implemented a relational database in Hyper SQL DB to store application data. Followed a 3 layer architecture.

PROJECTS cont

Piston Pump | Fusion360, KiCAD, C++, Atmel SAMD 2019 - 2022

- Designed and built a piston based liquid delivery device that exceeded ASTM class A standards for accuracy and delivery speed
- Designed touchscreen interface for human interaction, control, calibration, error recovery

City Winnipeg Database | Python, Qt, SQLite, Google Maps API, GeoPandas, Pandas 2021

- Combined multiple open data sources about the City of Winnipeg into a 3NF and BCNF normalized relational database
- Used Google Maps API to get asset geographic coordinates and used GeoPandas to measure distances and regional inclusions
- Created Windows, Linux cross platform GUI program in Python to run queries against SQLite DB

Fuzzy video content matcher | C/C++, Python, Ubuntu 2020 - 2022

Perceptual hashing to find similar videos. Implemented a python script on Linux to use Facebook TMK+PDQF algorithm

3D Printer | Fusion360, C++, RAMPS, Arduino Mega, RaspberryPi 2014 - 2017

Designed and built an FDM 3D printer with control and monitoring via web interface

Wireless controllable outlet hub | C/C++, HTML, CSS, JS, esp8266 2016

Lan/Wifi connected hub for controlling remote switches over 433mhz with cross platform embedded web page for control

EXTRACURRICULAR

University Drone Club | Git, C/C++, Python, OpenCV, Qgroundcontrol 2022 - 2023

- Created QR code reader to parse drone waypoints, and GPS boundary points to avoid
- Developed algorithm to avoid a Geo boundary and find a path to the next waypoint
- Developed system to build json files with updated Qgroundcontrol mission and upload the new mission via MAVLink
- Created AI subsystem to autonomously identify people, measure distance, and track movement from an airborne drone

University Dev Club Hackathon | Git, Python, OpenCV, Mediapipe 2023

Implemented gesture recognition on top of the MediaPipe ML library to create a hand tracking mouse in 12hrs

University Rocket club Avionics GUI Software | Git, C/C++, Qt, FreeRTOS, STMicro 2021 - 2022

Created cross platform desktop GUI application to allow avionics board configuration for general users

University Biomedical Engineering club Gaming Head Mouse | C/C++, Mbed OS 2021 - 2022

Created a head mounted gyroscopic joystick to allow a disabled gamer play with head movements

Magpie CTF Hackathon| Ghidra, IDA Free, Ollydbg, HTML, CSS, JS, C/C++, Go 2021

Participated in a capture the flag hackathon involving reverse engineering and penetration testing